Multi-Asset VaR and Expected Shortfall Analysis

Equity Option and Fixed Income Portfolio Risk

Executive Summary

Project Overview

This project presents a comprehensive multi-asset portfolio risk analysis focused on the measurement of Value-at-Risk (VaR) and Expected Shortfall (ES) across equity derivatives and fixed income portfolios.

The analysis evaluates and compares different quantitative risk methodologies, including:

- Delta-Normal (Asset-Normal) VaR,
- market beta mapping,
- Historical Simulation,
- fixed income cash-flow mapping,
- and portfolio-level risk aggregation.

The objective is to assess how different modeling assumptions impact downside risk estimation, tail-risk measurement and portfolio exposure under stressed market conditions.

The project combines:

- equity option risk modeling,
- fixed income risk analytics,
- yield curve risk estimation,
- FX-adjusted portfolio aggregation,
- and historical stress scenario analysis

within a realistic institutional portfolio management framework.

Part I: Equity Option Portfolio Risk

The equity option portfolio consisted of:

- a long position in 7,000 Nestlé (NESN) put options,
- and a short position in 2,000 Swiss Re (SREN) call options,

with each contract referencing 1,000 units of the underlying asset.

The analysis focused on estimating the portfolio's 1-day VaR(99%) and Expected Shortfall using multiple quantitative approaches.

1. Delta-Normal VaR (Asset-Normal Approach)

Metric	Value
Portfolio Volatility	CHF 3,657,708
VaR(99%,1-day)	CHF 8,509,102
ES(99%,1-day)	CHF 9,748,576

The portfolio risk was estimated by mapping option positions to their monetary deltas and assuming jointly normally distributed returns for the underlying assets.

The Delta-Normal framework provides a fast and computationally efficient approximation of market risk. However, due to the nonlinear payoff structure of options, the model tends to underestimate downside exposure during periods characterized by large market movements and fat-tail behavior.

Despite these limitations, the resulting VaR and ES estimates remain coherent with the assumptions underlying the variance-covariance framework.

2. SMI-Mapped VaR (Systematic Market Exposure)

Metric	Value
SMI-adjusted Exposure	CHF 1,349,877
VaR(99%,1-day)	CHF 3,140,283
ES(99%,1-day)	CHF 3,597,710

The portfolio was subsequently mapped to the Swiss Market Index (SMI) through beta exposure estimation in order to isolate the systematic component of market risk.

The materially lower VaR and ES values suggest that the mapping approach captures broad market exposure but fails to fully incorporate:

- option convexity,
- idiosyncratic exposure,
- and nonlinear portfolio dynamics.

While useful for measuring systematic market sensitivity, this methodology significantly understates the true downside risk of derivative portfolios under stressed market conditions.

3. Historical Simulation Approach

Metric	Value
1st Percentile Return	-51.50%
VaR(99%,1-day)	CHF 14,536,821
ES(99%,1-day)	CHF 17,813,658

The Historical Simulation framework generated 250 joint return scenarios for NESN and SREN using observed historical market movements while keeping implied volatility and risk-free rates constant.

The option portfolio was fully revalued under each simulated market scenario in order to derive empirical loss distributions.

Compared to parametric methodologies, Historical Simulation produced substantially higher VaR and ES estimates, reflecting:

- fat-tail behavior,
- volatility clustering,
- and the nonlinear characteristics of option payoffs.

The analysis confirms that Historical Simulation provides a more realistic representation of downside exposure for derivative portfolios because it avoids restrictive distributional assumptions and directly incorporates historical market stress events.

Part II: Fixed Income Portfolio Risk

The fixed income portfolio consisted of USD-denominated U.S. government bonds with different maturities and coupon structures.

The analysis focused on measuring portfolio exposure to:

- interest-rate movements,
- yield curve shifts,
- and fixed income valuation risk.

1. Bond Portfolio Pricing

Metric	Value
Total Market Value	USD 429,859,387

The market value of each Treasury bond was computed by discounting future cash flows using the zero-coupon term structure observed on April 25, 2025.

The resulting valuation framework allowed the construction of a full fixed income portfolio exposure model based on the underlying yield curve dynamics.

2. Fixed Income Delta-Normal VaR

Metric	Value
Portfolio Volatility	USD 978,748
VaR(99%,1-day)	USD 2,276,909
ES(99%,1-day)	USD 2,608,574

Risk estimation was performed using a cash-flow mapping methodology based on:

- maturity buckets,
- duration exposure,
- and the variance-covariance matrix of historical yield changes.

The Delta-Normal approach generated relatively moderate risk estimates under the assumption of stable interest-rate dynamics and normally distributed yield movements.

Although useful for benchmarking and portfolio comparison purposes, this framework may underestimate exposure during periods characterized by:

- abrupt yield curve shifts,
- market dislocations,
- and stressed liquidity conditions.

3. Historical Simulation Approach

Metric	Value
1st Percentile Return	-0.60%
VaR(99%,1-day)	USD 2,416,697
ES(99%,1-day)	USD 5,446,896

Historical Simulation was implemented by generating 250 hypothetical yield curve scenarios using observed historical changes across the term structure.

The fixed income portfolio was revalued under each simulated scenario in order to obtain empirical portfolio loss distributions.

The significantly higher Expected Shortfall relative to VaR highlights the presence of potentially severe downside losses under stressed yield curve environments.

The analysis confirms that Historical Simulation captures:

- nonlinear interest-rate risk,
- extreme market shocks,
- and tail-risk behavior

more effectively than traditional variance-covariance methodologies.

Part III: Total Fund Risk Aggregation

The final stage of the project focused on aggregating:

- the equity derivative portfolio,
- and the fixed income portfolio

into a single CHF-denominated multi-asset fund risk framework.

The analysis explicitly incorporated the USD-denominated bond exposure and the CHF/USD exchange rate.

1. Conservative Aggregation Estimate

Metric	Value
1st Percentile Return	-3.49%
VaR(99%,1-day)	CHF 15,090,501
ES(99%,1-day)	CHF 25,368,989

2. Historical Simulation Aggregation

Metric	Value
1st Percentile Return	-4.39%
VaR(99%,1-day)	CHF 16,835,500
ES(99%,1-day)	CHF 22,810,995

The empirical aggregation approach generated moderately higher portfolio risk estimates relative to the conservative framework.

Despite converting all positions into CHF, the overall impact of FX risk remained limited due to the relative stability of the CHF/USD exchange rate during the sample period.

The analysis further demonstrates that:

- diversification effects partially reduced aggregate portfolio risk,
- while historical simulation remained more effective in capturing tail-risk exposure under stressed market conditions.

Conclusion

This project demonstrates how different quantitative risk methodologies can produce materially different assessments of downside exposure across multi-asset portfolios.

By combining:

- equity option risk analytics,
- fixed income valuation techniques,
- volatility and yield curve modeling,
- historical stress testing,
- and portfolio-level aggregation procedures,

the analysis provides a realistic institutional framework for measuring and interpreting market risk.

The results highlight the importance of:

- model selection,
- distributional assumptions,
- nonlinear exposure treatment,
- FX-adjusted aggregation,
- and historical stress scenario construction

when evaluating portfolio downside risk and regulatory capital exposure.

The project ultimately illustrates how advanced quantitative methodologies can support more informed:

- portfolio management,
- market risk monitoring,
- and institutional risk reporting decisions.